

Which of the following enzymes are used in the synthesis of glucose from lactate?

- I. Lactate dehydrogenase
- II. Fructose-1,6-bisphosphatase
- III. Pyruvate kinase
- IV. Phosphoglycerate kinase
- V. Glyceraldehyde-3-phosphate dehydrogenase

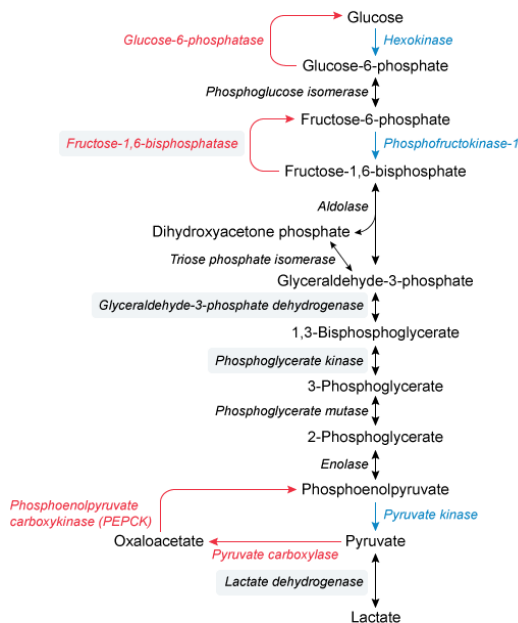
- ☐ A. I, II, and V only [30%]
- ☐ B. III, IV, and V only [7%]
- ✓ ☒ C. I, II, IV, and V only [50%]
- ☐ D. I, II, III, IV, and V [11%]

Correct

50% answered correctly

Explanation:

Gluconeogenesis shares several enzymes with glycolysis and has unique enzymes for bypass reactions



Blue = irreversible reactions of glycolysis.
Red = bypass reactions of gluconeogenesis.

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Many steps of metabolic pathways **operate near equilibrium** (ie, $\Delta G \approx 0$) at physiological conditions. These reactions are **reversible** and can switch between forward and reverse directions with only **small perturbations** to metabolite concentrations. In contrast, spontaneous reactions that operate **far from equilibrium** (ie, large, negative ΔG) are **irreversible** and are used in either the catabolic or the anabolic pathway, but not both.

Gluconeogenesis is the **anabolic** production of glucose from **non-carbohydrate precursors** such as **pyruvate** and **lactate**. These gluconeogenic precursors can also be produced from glucose through the catabolic process of glycolysis and fermentation, respectively. Most of the enzymes of gluconeogenesis are shared with glycolysis because most of the reactions in both pathways operate near equilibrium. For example, the reactions catalyzed by the enzymes **glyceraldehyde-3-phosphate dehydrogenase** (step 6 of glycolysis) and **phosphoglycerate kinase** (step 7 of glycolysis) operate in **both glycolysis and gluconeogenesis (Numbers IV and V)**; therefore, these enzymes are involved in production of glucose from lactate.

Lactate dehydrogenase is another reversible enzyme. In oxygen-poor tissue, lactate dehydrogenase uses pyruvate to oxidize NADH, producing **lactate** and NAD^+ in a process called **fermentation**. In oxygen-rich tissue, an excess of NAD^+ allows the **oxidation of lactate back to pyruvate**, which then continues through gluconeogenesis to make glucose (**Number I**).

In contrast, phosphorylation of fructose 6-phosphate by ATP (step 3 of glycolysis) has a **large, negative ΔG** and therefore is irreversible. Consequently, gluconeogenesis **cannot use the same enzyme** because phosphate transfer from fructose 1,6-bisphosphate to ADP (ie, the reverse reaction) has a large **positive ΔG** and is nonspontaneous. Therefore, gluconeogenesis uses a **bypass reaction** that produces fructose 6-phosphate from fructose 1,6-bisphosphate through simple hydrolysis, without any ATP or ADP involvement at all. The enzyme that catalyzes this hydrolysis reaction is **fructose-1,6-bisphosphatase (Number II)**.

(Number III) Pyruvate kinase catalyzes a glycolytic reaction with a large, negative ΔG . Its reverse reaction has a large, positive ΔG , is nonspontaneous under physiological conditions, and therefore *is not used* in gluconeogenesis to produce glucose from lactate.

Educational objective:

Glucose is synthesized from precursors (eg, pyruvate, lactate) through gluconeogenesis. Gluconeogenesis shares seven enzymes with glycolysis; however, the irreversible steps of glycolysis are bypassed by distinct enzyme-catalyzed reactions unique to gluconeogenesis.

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Subject:
Biochemistry

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